

Build Smarter AI Apps: Empower LLMs with LangChain

Module Cheat Sheet: Introduction to LangChain in GenAI

Package/Method	Description	Code Example
WatsonxLLM	A class from the <code>ibm_watson_machine_learning.foundation_models.extensions.langchain</code> module that creates a LangChain compatible wrapper around IBM's watsonx.ai models.	<pre>from ibm_watsonx_ai.foundation_models import Model from ibm_watson_machine_learning.foundation_models.extensions.langchain import WatsonxLLM model_id = 'mistralai/mixtral-8x7b-instruct-v0.1' parameters = { GenParams.MAX_NEW_TOKENS: 256, GenParams.TEMPERATURE: 0.2, } credentials = {"url": "https://us-south.ml.cloud.ibm.com", "apikey": "sk-xxxxx"} project_id = "skills-network" model = ModelInference(model_id=model_id, params=parameters, credentials=credentials, project_id=project_id) mixtral_llm = WatsonxLLM(model=model) response = mixtral_llm.invoke("Who is the president of the United States?")</pre>
Message Types	Different types of messages that chat models can use to provide context and control the conversation. The most common message types are <code>SystemMessage</code> , <code>HumanMessage</code> , and <code>AIMessage</code> .	<pre>from langchain_core.messages import HumanMessage, SystemMessage msg = mixtral_llm.invoke([SystemMessage(content="You are a helpful assistant."), HumanMessage(content="I enjoy mystery novels.")])</pre>
PromptTemplate	A class from the <code>langchain_core.prompts</code> module that helps format prompts with variables. These templates allow you to define a consistent format while leaving placeholders for variables that change with each use case.	<pre>from langchain_core.prompts import PromptTemplate prompt = PromptTemplate.from_template('Tell me a {adjective} joke about {topic}.') input_ = {"adjective": "funny", "topic": "cats"} formatted_prompt = prompt.invoke(input_)</pre>
ChatPromptTemplate	A class from the <code>langchain_core.prompts</code> module that formats a list of chat messages with variables. These templates consist of a list of message templates themselves.	<pre>from langchain_core.prompts import ChatPromptTemplate prompt = ChatPromptTemplate.from_messages([("system", "You are a helpful assistant."), ("user", "Tell me a joke about {topic}."),]) input_ = {"topic": "cats"} formatted_messages = prompt.invoke(input_)</pre>
MessagesPlaceholder	A placeholder that allows you to add a list of messages to a specific spot in a <code>ChatPromptTemplate</code> . This capability is useful when you want the user to pass in a list of messages you would slot into a particular spot.	<pre>from langchain_core.prompts import MessagesPlaceholder from langchain_core.messages import HumanMessage, SystemMessage prompt = ChatPromptTemplate.from_messages([("system", "You are a helpful assistant."), MessagesPlaceholder("history"), ("user", "Tell me a joke about {topic}."),])</pre>

		<pre> MessagesPlaceholder("msgs")]) input_ = {"msgs": [HumanMessage(content formatted_messages = prompt.invoke(inp </pre>
JsonOutputParser	A parser that allows users to specify an arbitrary JSON schema and query LLMs for outputs that conform to that schema. A parser is useful for obtaining structured data from LLMs.	<pre> from langchain_core.output_parsers import from langchain_core.pydantic_v1 import class Joke(BaseModel): setup: str = Field(description="que punchline: str = Field(description= output_parser = JsonOutputParser(pydant format_instructions = output_parser.get prompt = PromptTemplate(template="Answer the user query.\n{ input_variables=["query"], partial_variables={"format_instruct) chain = prompt mixtral_llm output_p </pre>
CommaSeparatedListOutputParser	A parser used to return a list of comma-separated items. This parser converts the LLM's response into a Python list.	<pre> from langchain.output_parsers import Co output_parser = CommaSeparatedListOutpu format_instructions = output_parser.get prompt = PromptTemplate(template="Answer the user query. { input_variables=["subject"], partial_variables={"format_instruct) chain = prompt mixtral_llm output_p result = chain.invoke({"subject": "ice </pre>
Document	A class from the langchain_core.documents module that contains information about some data. This class has the following two attributes: page_content (the content of the document) and metadata (arbitrary metadata associated with the document).	<pre> from langchain_core.documents import Do doc = Document(page_content="Python is an interpr Python's design phi metadata={ 'my_document_id' : 234234, 'my_document_source' : "About f 'my_document_create_time' : 168 }) </pre>
PyPDFLoader	A document loader from the langchain_community.document_loaders that loads PDFs into Document objects. You can use this document loader to extract text content from PDF files.	<pre> from langchain_community.document_loade loader = PyPDFLoader("path/to/document. documents = loader.load() </pre>

WebBaseLoader	A document loader from the langchain_community.document_loaders that loads content from websites into Document objects. You can use this document loader to extract text content from web pages.	<pre> from langchain_community.document_loaders import WebBaseLoader loader = WebBaseLoader(["https://python.org"]) web_data = loader.load() </pre>
CharacterTextSplitter	A text splitter from langchain.text_splitter that splits text into chunks based on characters. This splitter is useful for breaking long documents into smaller, more manageable chunks for processing with LLMs.	<pre> from langchain.text_splitter import CharacterTextSplitter text_splitter = CharacterTextSplitter(chunk_size=200, # Maximum size of chunk_overlap=20, # Number of char separator="\n" # Character to split on) chunks = text_splitter.split_documents(docs) </pre>
RecursiveCharacterTextSplitter	A text splitter from langchain.text_splitter that splits text recursively based on a list of separators. This splitter tries to split on the first separator, then the second separator, and any subsequent separators, until the chunks of text attain the specified size.	<pre> from langchain.text_splitter import RecursiveCharacterTextSplitter text_splitter = RecursiveCharacterTextSplitter(chunk_size=500, chunk_overlap=50, separators=["\n\n", "\n", ". ", " ", ""]) chunks = text_splitter.split_documents(docs) </pre>
WatsonxEmbeddings	A class from langchain_ibm that creates embeddings (vector representations) of text using IBM's watsonx.ai embedding models. You can use these embeddings for semantic search and other vector-based operations.	<pre> from langchain_ibm import WatsonxEmbeddings from ibm_watsonx_ai.metanames import EmbedTextParamsMetaNames embed_params = { EmbedTextParamsMetaNames.TRUNCATE_LENGTH: 100, EmbedTextParamsMetaNames.RETURN_OPTIONS: "EMBEDDINGS" } watsonx_embedding = WatsonxEmbeddings(model_id="ibm/slate-125m-english-rtr", url="https://us-south.ml.cloud.ibm.com", project_id="skills-network", params=embed_params,) </pre>
Chroma	A vector store from langchain.vectorstores that stores embeddings and provides methods for similarity search. You can use Chroma for storing and retrieving documents based on semantic similarity.	<pre> from langchain.vectorstores import Chroma // Create a vector store from documents docsearch = Chroma.from_documents(chunked_documents, embedding) // Perform a similarity search query = "Langchain" docs = docsearch.similarity_search(query) </pre>

Retrievers	Interfaces that return documents given an unstructured query. Retrievers accept a string query as input and return a list of Document objects as output. You can use vector stores as the backbone of a retriever.	<pre># Convert a vector store to a retriever retriever = docsearch.as_retriever() // Retrieve documents docs = retriever.invoke("Langchain")</pre>
ParentDocumentRetriever	A retriever from langchain.retrievers that splits documents into small chunks for embedding but returns the parent documents during retrieval. This retriever balances accurate embeddings with context preservation.	<pre>from langchain.retrievers import Parent from langchain.storage import InMemoryStore parent_splitter = CharacterTextSplitter(chunk_size=1000, chunk_overlap=0) child_splitter = CharacterTextSplitter(chunk_size=500, chunk_overlap=0) vectorstore = Chroma(collection_name="split_parents", embedding_function=watsonx_embedder, persist_directory='./chroma_db',) store = InMemoryStore() retriever = ParentDocumentRetriever(vectorstore=vectorstore, docstore=store, child_splitter=child_splitter, parent_splitter=parent_splitter,) retriever.add_documents(documents) retrieved_docs = retriever.invoke("Langchain")</pre>
RetrievalQA	A chain from langchain.chains that answers questions based on retrieved documents. The RetrievalQA chain combines a retriever with an LLM to generate answers based on the retrieved context.	<pre>from langchain.chains import RetrievalQA qa = RetrievalQA.from_chain_type(llm=mixtral_llm, chain_type="stuff", retriever=docsearch.as_retriever(), return_source_documents=False) query = "what is this paper discussing?" answer = qa.invoke(query)</pre>
ChatMessageHistory	A lightweight wrapper from langchain.memory that provides convenient methods for saving HumanMessages, AIMessages, and then fetching them all. You can use the ChatMessageHistory wrapper to maintain conversation history.	<pre>from langchain.memory import ChatMessageHistory history = ChatMessageHistory() history.add_ai_message("hi!") history.add_user_message("what is the capital of France?") // Access the messages history.messages // Generate a response using the history ai_response = mixtral_llm.invoke(history.messages)</pre>
ConversationBufferMemory	A memory module from langchain.memory that allows for the storage of messages and conversation history. You can use this memory module conversation chains to maintain context across multiple interactions.	<pre>from langchain.memory import ConversationBufferMemory from langchain.chains import ConversationChain conversation = ConversationChain(llm=mixtral_llm, memory=ConversationBufferMemory()</pre>

		<pre> verbose=True, memory=ConversationBufferMemory()) response = conversation.invoke(input="") </pre>
LLMChain	A basic chain from langchain.chains that combines a prompt template with an LLM. It's the simplest form of chain in LangChain.	<pre> from langchain.chains import LLMChain template = """Your job is to come up with a recipe for a meal based on the location: {location} YOUR RESPONSE: """ prompt_template = PromptTemplate(template=template, input_variables=["location"]) location_chain = LLMChain(llm=mixtral_llm, prompt=prompt_template, output_key='meal') result = location_chain.invoke(input='New York') </pre>
SequentialChain	A chain from langchain.chains that combines multiple chains in sequence, where the output of one chain becomes the input for the next chain. SequentialChain is useful for multi-step processing.	<pre> from langchain.chains import SequentialChain // First chain - gets a meal based on location location_chain = LLMChain(llm=mixtral_llm, prompt=location_prompt_template, output_key='meal') // Second chain - gets a recipe based on meal dish_chain = LLMChain(llm=mixtral_llm, prompt=dish_prompt_template, output_key='recipe') // Third chain - estimates cooking time recipe_chain = LLMChain(llm=mixtral_llm, prompt=recipe_prompt_template, output_key='time') // Combine into sequential chain overall_chain = SequentialChain(chains=[location_chain, dish_chain, recipe_chain], input_variables=['location'], output_variables=['meal', 'recipe', 'time'], verbose=True) </pre>
RunnablePassthrough	A component from langchain_core.runnables that allows function chaining to use the 'assign' method, enabling structured multi-step processing.	<pre> from langchain_core.runnables import RunnableAssign // Create each individual chain with the LLM location_chain_lcel = (PromptTemplate.from_template(location_prompt_template), mixtral_llm, StrOutputParser()) dish_chain_lcel = (PromptTemplate.from_template(dish_prompt_template), mixtral_llm, StrOutputParser()) time_chain_lcel = (PromptTemplate.from_template(time_prompt_template), mixtral_llm, StrOutputParser()) // Combine chains using RunnableAssign overall_chain_lcel = location_chain_lcel RunnableAssign({"meal": location_chain_lcel.output_key, "recipe": dish_chain_lcel.output_key, "time": recipe_chain_lcel.output_key}) </pre>

		<pre> StrOutputParser()) overall_chain_lcel = (RunnablePassthrough.assign(meal=lan RunnablePassthrough.assign(recipe RunnablePassthrough.assign(time=]) // Run the chain result = overall_chain_lcel.invoke({"lc pprint(result) </pre>
Tool	A class from langchain_core.tools that represents an interface that an agent, chain, or LLM can use to interact with the world. Tools perform specific tasks like calculations and data retrieval.	<pre> from langchain_core.tools import Tool from langchain_experimental.utilities import PythonREPL python_repl = PythonREPL() python_calculator = Tool(name="Python Calculator", func=python_repl.run, description="Useful for when you need to perform calculations") result = python_calculator.invoke("a + b = ?") </pre>
@tool decorator	A decorator from langchain.tools that simplifies the creation of custom tools. This tool automatically converts a function into a Tool object.	<pre> from langchain.tools import tool @tool def search_weather(location: str): """Search for the current weather in a given location. # In a real application, this function would make an API call. return f"The weather in {location} is sunny." </pre>
create_react_agent	A function from langchain.agents that creates an agent following the ReAct (Reasoning + Acting) framework. This function takes an LLM, a list of tools, and a prompt template as input and returns an agent that can reason and select tools to accomplish tasks.	<pre> from langchain.agents import create_react_agent agent = create_react_agent(llm=mixtral_llm, tools=tools, prompt=prompt) </pre>
AgentExecutor	A class from langchain.agents that manages the execution flow of an agent. This class handles the orchestration between the agent's reasoning and the actual tool execution.	<pre> from langchain.agents import AgentExecutor agent_executor = AgentExecutor(agent=agent, tools=tools, verbose=True, handle_parsing_errors=True) result = agent_executor.invoke({"input": "What is the weather in New York?"}) </pre>

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Skills Network